
HOMEWORK DAY 22 – *Laplace Transform – Solving ODEs §3.2*

1. Find the inverse Laplace transform of

(a) $F(s) = \frac{2\pi}{s+2}$

(b) $F(s) = \frac{2}{s(s+2)}$ (Hint: use partial fraction decomposition)

2. Solve the following initial value problem

$$x' = t + 3, \quad x(0) = 2$$

(a) Using a method from Chapter 1.

(b) Using the Laplace Transform, using the table of transforms we have developed so far.

3. p145: 9f

4. p145: 9g

5. p156: 3b

6. p157: 6d

7. p157: 6e

HOMEWORK DAY 23 – *Laplace Transform – Shifts*

8. Find the following Laplace transforms.

(a) $\mathcal{L}[\cos(7t)]$

(b) $\mathcal{L}[e^{-3t} \sin(2t)]$

(c) $\mathcal{L}[8 + 2e^{-t} + te^{2t}]$

9. Find the following inverse Laplace transforms.

(a) $\mathcal{L}^{-1} \left[\frac{2}{(s-5)^2} \right]$

(b) $\mathcal{L}^{-1} \left[\frac{s}{(s-5)^2} \right]$

$$(c) \quad \mathcal{L}^{-1} \left[\frac{3 - 2s}{s^2 + 2s + 10} \right] \quad (\text{p156: 3b})$$

$$(d) \quad \mathcal{L}^{-1} \left[\frac{5s}{(s - 3)^2 + 4} \right] \quad (\text{p156: 3h})$$

Use the Laplace Transform to solve the following initial value problems.

10. $x'' + 9x = \sin(2t)$, $x(0) = x'(0) = 0$ (p157: 6h)

11. $x'' - 2x' + 5x = 0$, $x(0) = 0$, $x'(0) = 1$